

8(a)	energy gained by an electron with charge $1.6 \times 10^{-19} \text{ C}$	B1
	when accelerated / moved <u>through a p.d. of one volt</u>	B1
(b)(i)	energy of photon = $(60 \times 10^3) \times (1.6 \times 10^{-19})$	C1
	wavelength = $(6.63 \times 10^{-34}) \times (3.0 \times 10^8) / 9.6 \times 10^{-15}$	C1
	= 2.1×10^{-11} or $2.07 \times 10^{-11} \text{ m}$	A1
(b)(ii)	(incident) electrons encountering target (anode) lose energy to give several X-ray photons of lower energy / higher wavelength than (b)(i) .	B1
(c)(i)	energy = $10^{4.84} - 10^{4.06}$	C1
	= 57700 eV = 57.7 keV	A1
(c)(ii)	Use either the proportionality constant from the Moseley for each element or compare the ratios of Z and $\sqrt{E}$.	M1
	Ratios involve at least 3 elements	M1
	Conclusion with valid reason	A1
(c)(iii)	K_β photon is created by an energy drop that is larger than the K_α photon	B1
(d)	energy is produced as heat in the target	B1
	tungsten has a high melting point (so will not melt)	B1
	OR tungsten has a high proton number	B1
	greater probability of collisions between electrons and the large nucleus	B1
(e)(i)(1)	penetrating power = $\ln 2 / 0.528$ = 1.31 cm	C1
(e)(i)(2)	$I/I_0 = 0.40 = e^{-0.528x}$	M1
	x = 1.74 cm	A1

(e)(ii)	$0.60 = e^{-\mu(3.87)}$ gives $\mu = 0.132 \text{ cm}^{-1}$	A1
(e)(iii)	Different coefficient for different body material	B1
	Coefficient affects brightness / intensity of parts	B1